Reporting Regulation, Market Liquidity, and Policy Diffusion

Abstract

We examine reporting regulation in a global sample of 90 jurisdictions over 2000-2025. The paper has two parts. First, we aggregate stock liquidity within countries and compare changes around national implementation with changes in comparison countries. Seventeen regulation families are considered; 15 support first-year comparisons and nine support a longer window. The first-year spread estimate is -0.029 under equal family weights and +0.021 under market-cap weights, with both intervals spanning zero. Second, we examine whether competition, learning, emulation and external finance are associated with national adoption. We report both family-weighting schemes on matched samples. Only the external-finance association under market-cap family weights survives adjustment across the eight planned comparisons. The learning comparison has insufficient observations. None of the weighted associations survives the broader matrix correction. The evidence concerns national market exposure; company-specific reporting obligations are examined separately.

Introduction

Reporting rules can make companies easier to compare and reduce the information disadvantage faced by investors. They may therefore lower the cost of trading. The benefit need not be the same for every rule: some disclosures reach investors, while others are sent to tax authorities, and compliance costs or limited enforcement can offset information gains. We ask how stock-market liquidity changes around national regulation and how the policies spread across countries.

We extend the international disclosure setting in Krueger et al. (2023) beyond mandatory ESG reporting to 17 regulation families. Our first part studies market liquidity at the country level. Our second part follows Koning, Mertens and Roosenboom (2018) in examining competition, learning, emulation and external finance as explanations for adoption. Public disclosure, confidential tax reporting and regulatory institutions are kept distinct, because their information channels differ.

1. Liquidity

Datastream provides 1,055,142 stock-year observations for 99,100 vendor company records in 90 jurisdictions over 2000-2025. We form 2,050 country-year observations. The sample is broad, but is not a census of every listed company. We measure quoted bid-ask spreads, the price response to trading volume, and the share of daily returns equal to zero. Each measure requires 120 valid daily observations; higher values indicate lower liquidity. Spreads and price impact enter in logs and are winsorised by year at the first and ninety-ninth percentiles.

Table 1 describes the regulation families. For each family and adoption cohort, we hold the company sample fixed, average company liquidity changes within each country, and compare adopting countries with documented comparison countries. Already-adopting countries are excluded from the comparison group, and unknown histories are not treated as nonadoption. The first-year estimates cover 48 adopting jurisdictions across 15 families; the longer window covers 36 across nine. The first comparison is year 0 against year -1. The longer window averages years 0 to 2 against year -1, with fixed observations from -3 through +2.

We weight country averages by GDP in the year before adoption. Within each family, adoption cohorts receive weight proportional to sampled adopting-country GDP. We then combine families in two ways: each family receives the same weight, or receives weight proportional to sampled company market value before implementation. Both schemes use the same family estimates. Countries need at least 30 companies with complete outcomes, and each comparison needs three comparison countries. Standard errors allow observations within a country to move together. National rules can also arrive in common legal waves, which limits causal inference.

Table 2 asks whether liquidity improves across the regulation families. First-year log spreads are -0.029 under equal family weights and +0.021 under market-cap weights; neither is precise. Price impact and unchanged returns also depend on the weighting scheme. Only the equal-family unchanged-return result survives correction across the 12 principal liquidity comparisons. Figure 1 shows the family estimates and weights. Confidential tax country reporting receives 33.2% of market-cap weight and auditor reporting 9.9%. Figure 2 shows differences before implementation, so the subsequent changes cannot be attributed to regulation alone. The evidence does not establish a common liquidity gain across these policies.

2. Policy diffusion

We next examine why countries adopt. The adoption panel contains documented histories for 90 jurisdictions in the global market sample. We include collective implementation waves and examine independent national decisions separately. Each country remains at risk until its recorded implementation year. Countries without a documented history do not enter as nonadopters. Foreign peers can include other jurisdictions with documented histories. The market-value weights are available for the 15 liquidity-supported families; both weighting schemes use that common eligible family set, subject to mechanism-specific data availability.

Following Koning, Mertens and Roosenboom (2018), competition measures earlier adoption among countries with similar export markets. Learning compares earlier economic outcomes among neighbouring adopters and nonadopters. Emulation measures adoption among language peers. External finance measures IMF credit and aid relative to GDP. Signals precede the adoption decision. Separate logit models compare countries within the same family and year. Equal-family weights give each represented family the same total weight; market-cap weights use the same pre-implementation family values as the liquidity analysis, renormalised over the represented families. Each pair of weighted models uses identical observations and predictor units, with country-clustered uncertainty.

Table 3 reports the four mechanisms under both weights. The external-finance coefficients are negative: equal family -1.111 (adjusted p=0.142); market-cap family -0.967 (adjusted p=0.018). Only the external-finance association under market-cap family weights survives adjustment across the eight planned comparisons. The learning comparison has insufficient observations. None of the weighted associations survives the broader matrix correction. Some unweighted adoption-status models show associations, particularly for public bank reporting; the pooled weighted status models do not survive the broader correction. Adoption status describes an established regime, while timing concerns the adoption decision. The appendix reports both, along with controls, independent decisions, liquidity-based learning and combined mechanisms. These associations do not establish that one country caused another to adopt.

References

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied working paper.

Koning, M., G. Mertens and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies. Supplied article.

Policy chronology, official source links, legal-scope reviews, complete estimates and implementation sensitivities: https://hunternbh.github.io/policy-diffusion/ . Licensed Datastream observations remain local.

Table 1. International policy coverage

Regulation family	Treated countries	Supported cohort years	Adoption cohorts	Market-cap weight	Longer window
Audit institutions / rotation	4	2004-2007	3	2.6%	Yes
Auditor reporting	19	2012-2016	2	9.9%	Yes
Beneficial ownership	11	2016-2020	5	10.3%	Yes
Public bank country reports	15	2014	1	6.5%	Yes
Confidential tax country reports	42	2016-2020	5	33.2%	Yes
Public multinational country reports	1	2023	1	<0.1%	No
Climate / emissions disclosure	2	2013-2023	2	2.0%	Yes
Sectoral climate rules	-	-	-	-	No
Digital-asset regulation	2	2018-2020	2	0.5%	Yes
Nonfinancial reporting	23	2002-2019	10	10.7%	Yes
Payments to governments	5	2014-2015	2	6.4%	No
Corporate governance	6	2001-2004	4	1.2%	No
IFRS financial reporting	21	2005-2011	6	6.5%	No
Supply-chain reporting	5	2012-2022	5	9.5%	Yes
Financial-focus sustainability	3	2024-2025	2	0.5%	No
Sustainability frameworks	-	-	-	-	No
Electronic financial reporting	1	2007	1	0.2%	No

First-year GDP-weighted specification. Only supported implementation cohorts through 2025 appear; future schedules are excluded. Counts refer to national policy exposure, not verified legally covered companies. No means insufficient longer-window support, not a zero effect. Public bank phases enter once. Audit institutions/rotation and governance frameworks include different policy subtypes. Nonfinancial dates are reviewed reporting phases, not a harmonised claim to the earliest E, S and G duties in every country.

Scope and timing

The calendar records the implemented policy phase supported by the source register. It does not claim the first existence of every reporting obligation. E-only climate reporting is distinct from comprehensive nonfinancial reporting; public and confidential country reports have different audiences. Historical thresholds, subsidiaries and foreign-listing exemptions are not fully observed. Unknown country histories remain outside the comparison set. The two unsupported families remain in the inventory without an estimated effect.

Table 2. International liquidity comparisons

Window and outcome	Equal family Estimate [interval]	Market-cap family Estimate [interval]
First year: 15 families Bid-ask spread (log)	-0.029 [-0.068, +0.011]	+0.021 [-0.021, +0.063]
First year: 15 families Price impact (log)	-0.045 [-0.087, -0.003]	+0.019 [-0.050, +0.087]
First year: 15 families Unchanged returns (fraction)	-0.0082 [-0.0112, -0.0052]	-0.0027 [-0.0065, +0.0012]
Years 0-2: 9 families Bid-ask spread (log)	-0.017 [-0.078, +0.044]	+0.045 [-0.035, +0.125]
Years 0-2: 9 families Price impact (log)	-0.013 [-0.081, +0.054]	+0.096 [-0.024, +0.216]
Years 0-2: 9 families Unchanged returns (fraction)	-0.0006 [-0.0064, +0.0053]	+0.0070 [-0.0070, +0.0210]

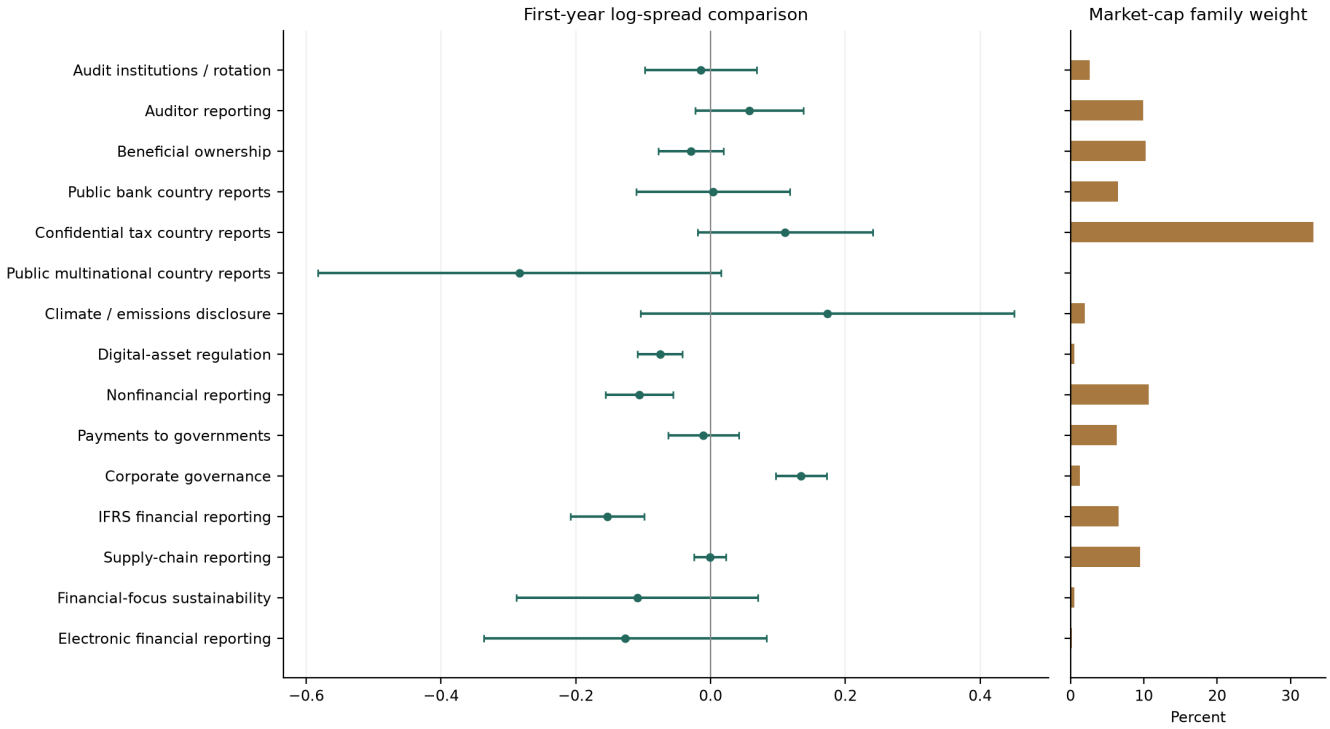
GDP weights within regulation families. Brackets are 95% country-clustered conditional intervals; they do not capture all uncertainty from collective policy waves, scope errors or endogenous implementation. Higher outcomes indicate lower liquidity. Log outcomes and fractions are not percentage changes. Horizons have different family and company support. All outcomes and both weights are reported.

Table 3. Country adoption and policy diffusion

Mechanism	Rows / countries / families	Equal family Coefficient [adjusted p]	Market-cap family Coefficient [adjusted p]
Competition	375 / 76 / 8	+0.346 [1.000]	+0.044 [1.000]
Learning	41 / 25 / 2	Not estimated	Not estimated
Emulation	419 / 53 / 12	+0.361 [1.000]	+0.215 [1.000]
External finance	154 / 30 / 7	-1.111 [0.142]	-0.967 [0.018]

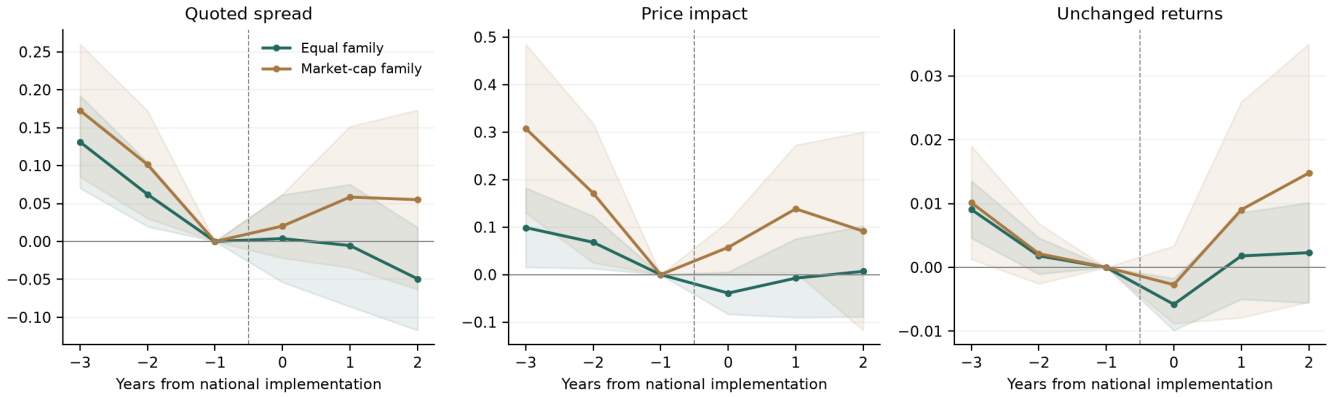
Separate weighted timing logit models with family-year fixed effects. Each pair uses identical country-family-year observations. Each family has equal total likelihood weight or its pre-implementation sampled market-cap mass, divided by usable rows. Country-clustered score-sandwich errors with t reference. Brackets report Holm-adjusted p-values across eight principal coefficients. Positive coefficients indicate a positive association with adoption. Continuous predictors are scaled by their common unweighted sample standard deviation; learning is binary. Wider matrix corrections and all unsupported models remain in the appendix.

Figure 1. International family estimates and weights



First-year national policy-exposure comparisons for all 15 supported families, GDP weighted within families. Bars show country-clustered conditional intervals. Right panel: sampled treated-company market capitalisation before implementation.

Figure 2. Liquidity around national implementation



Nine families with fixed -3 through +2 company support. Year -1 is the reference; year 0 is recorded national implementation. Shading shows pointwise country-clustered conditional intervals. Pre-implementation differences and collective legal waves limit a causal interpretation. This sample differs from the first-year sample.